Quantification: 872_wt_hp_z03
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	10
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2407.56	1386	3355.46	px^2
major_axis_length_px	lacuna shape	77.8702	69.6134	37.3439	px
minor_axis_length_px	lacuna shape	33.294	26.717	24.4483	px
aspect_ratio	lacuna shape	2.7332	2.443	1.108	unitless
eccentricity	lacuna shape	0.9002	0.9124	0.0603	unitless
solidity	lacuna shape	0.8717	0.8917	0.0457	unitless
roots_count	headline	8	7	4.1899	count
ring_length_r30_px	headline	326.7	293.5	115.532	px
ring_length_r60_px	reported	862.2	800.5	275.476	px
owned_length_px	ownership-dependent	1450.34	1173.16	1300.96	px
edge_count	ownership-dependent	90.1	72	83.3006	count
mean_edge_length_px	ownership-dependent	16.8145	16.1302	2.1219	px
perimeter_px	normalised	233.286	186.409	208.649	px
ring_area_r30_px2	normalised	8087.5	7821.5	3024.09	px^2
ring_area_r60_px2	normalised	21031.9	20698.5	5068.55	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0424774	0.0355188	0.0150842	px^-1
ring_density_r60	normalised	0.0418037	0.0385162	0.0128541	px^-1
roots_per_100px_perimeter	normalised	4.1962	4.2353	1.9337	per 100 px
ring_attached_length_r30_px	network v2	281	248	125.269	px
ring_attached_length_r60_px	network v2	579.8	480	358.09	px
ring_length_w_r30_px	network v2	370.339	333.957	133.115	px
ring_length_w_r60_px	network v2	973.973	901.595	314.464	px
sholl_crossings_r10	network v2	7.6	6.5	4.115	count
sholl_crossings_r20	network v2	8	7.5	3.3993	count
sholl_crossings_r30	network v2	11.6	10.5	3.9777	count
loop_count	loops, report only	9.7	4	10.9853	count
loop_edge_share	loops, report only	0.2696	0.2705	0.2079	share
ring_width_mean_r30_px	width	7.0219	6.7443	0.9486	px

Per field
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measure	value	units
analysed_area_px2	999693	px^2
total_skeleton_length_px	27705	px
canalicular_length_density_per_px	0.0277135	px^-1
junction_count	748	count
junction_density_per_px2	0.00074823	px^-2
lacuna_count	14	count

lacunae_per_px2	1.4e-05	px^-2
skeleton_component_count	389	count
mean_component_length_px	71.2211	px
median_component_length_px	16	px
field_length_density_w_per_px	0.0311523	px^-1
field_density_without_flagged_per_px	0.0273526	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	498	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0277135	px^-1
field_length_density_with_unattached_per_px	0.0282117	px^-1
loop_count	135	count
loop_edge_share	0.2665	share
edge_count	1861	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.2304	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25119	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	794	69.6134	15.6156	4.4579	0.9745	0.834	-1.442	210.909	115.354	False
2	420	36.8904	15.1005	2.443	0.9124	0.8955	-0.9718	230.283	755.805	False
3	1078	83.4971	17.8088	4.6885	0.977	0.8021	-1.3939	282.36	422.258	False
4	2506	64.9879	57.5952	1.1284	0.4632	0.8735	0.3596	332.631	998.485	True
5	2428	74.728	42.3101	1.7662	0.8243	0.9324	-1.5637	398.271	285.478	False
6	2726	99.126	35.5862	2.7855	0.9333	0.9172	-1.4054	396.683	51.522	True
7	1386	67.6197	26.7997	2.5232	0.9181	0.9053	-1.5154	396.681	828.755	False
8	1506	75.3481	26.717	2.8202	0.935	0.8917	-1.4312	511.238	191.668	False
9	1101	57.7118	24.8895	2.3187	0.9022	0.9137	-1.3389	569.522	707.45	False
10	568	28.9763	25.6237	1.1308	0.4669	0.9578	-0.8322	590.484	11.2324	True
11	1731	64.4432	36.6995	1.756	0.822	0.8469	-1.4227	651.578	232.505	False
12	11224	170.98	93.705	1.8247	0.8364	0.824	-1.4075	794.88	452.864	False
13	21222	397.246	71.5243	5.554	0.9837	0.8315	1.551	815.409	171.319	True
14								270.839	634.653	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	242	532	253.024	12	21.0854	6.4399
2	6	192	581	192.066	13	14.7743	5.808
3	5	230	622	291.966	17	17.1744	6.5575
4	6	272	717	1801.46	99	18.1966	6.2067
5	9	312	904	374.492	25	14.9797	7.7029

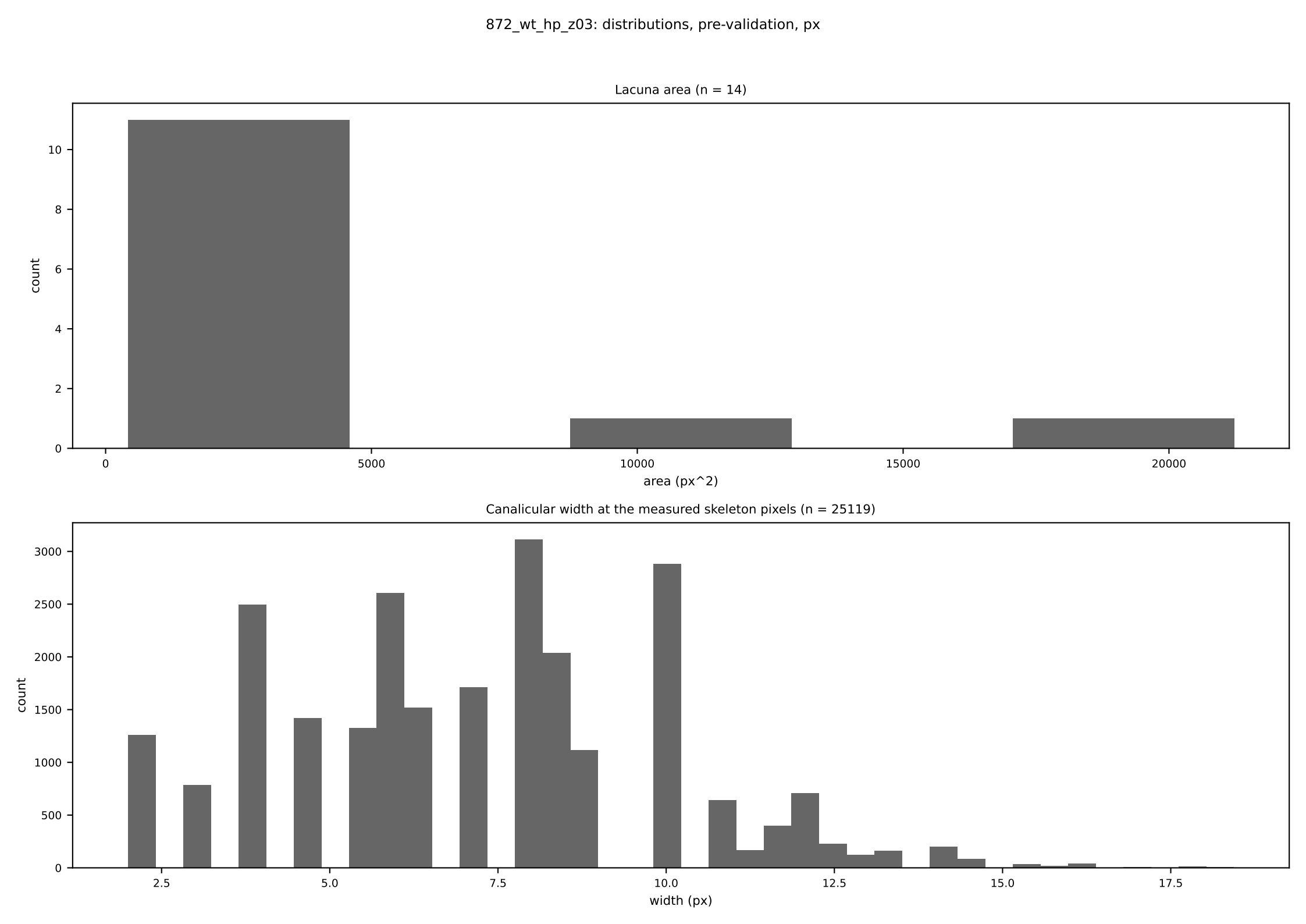
6	4	197	360	124.083	5	24.8167	5.7427
7	8	407	1143	2334.7	142	16.4415	6.7143
8	4	275	680	462.375	23	20.1032	6.7744
9	11	470	1100	2802.77	175	16.0158	8.3097
10	3	133	460	703.115	36	19.531	8.7365
11	14	525	1315	3796.33	245	15.4952	8.0788
12	15	395	1048	1883.94	119	15.8314	5.765
13	10	468	939	873.5	56	15.5982	5.3983
14	3	219	697	2111.78	130	16.2445	8.0687

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	165.782	7287	20197	1	1	0.0332098	0.0263406	3.016
2	97.0538	5433	16006	1	1	0.0353396	0.0362989	6.1821
3	229.8	8292	22152	1	1	0.0277376	0.0280787	2.1758
4	283.664	6163	14859	0.6902	0.6384	0.0441344	0.0482536	2.1152
5	223.179	8740	22193	1	1	0.0356979	0.0407336	4.0326
6	253.279	8248	19209	0.8394	0.7603	0.0238846	0.0187412	1.5793
7	180.267	7666	20955	1	1	0.0530916	0.0545455	4.4379
8	192.551	7977	20442	1	1	0.0344741	0.0332648	2.0774
9	151.782	6950	19530	1	1	0.0676259	0.0563236	7.2472
10	92.7696	3398	9619	0.6214	0.5799	0.0391407	0.047822	3.2338
11	219.522	8093	21078	1	1	0.0648709	0.0623873	6.3775
12	807.08	15857	33469	1	1	0.0249101	0.0313126	1.8586
13	1086.49	23683	45433	0.8953	0.8583	0.019761	0.0206678	0.9204
14	65.8406	4580	14297	1	1	0.0478166	0.0487515	4.5565

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	199	247	265.267	589.416	6	4	8
2	129	149	226.179	679.11	5	5	8
3	193	318	256.652	681.044	5	4	8
4	222	589	303.693	815.411	7	9	7
5	237	495	351.007	1006.74	5	11	13
6	100	139	210.924	391.948	3	9	7
7	317	904	471.546	1301.77	7	10	15
8	259	465	316.907	790.394	7	6	11
9	467	1040	541.132	1250.13	9	14	16
10	129	392	152.054	517.919	3	3	6
11	517	1227	605.558	1518.08	10	11	19
12	306	553	418.836	1126.58	18	8	10
13	341	554	499.291	1022.31	12	13	8
14	186	400	250.309	796.453	4	7	8


872_wt_hp_z03: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
0
2
4
6
8
10
0
5000
10000
15000
20000
Canalicular width at the measured skeleton pixels (n = 25119)
count
width (px)
0
500
1000
1500
2000
2500
3000
2.5
5.0
7.5
10.0
12.5
15.0
17.5